Assignment 1

Q1. [2+2 Marks] Select if the following tasks will require DDL(Data Definition Language) or DML(Data Manipulation Language).
Explain your answer briefly (at max 2 sentences) with mention to schema and state:

a. Rename the column called “StudentID” into “SID”
b. Change the “class time” for section 7 to “01:00 PM - 2:05 PM”

a) Ans: DDL, changes schema, not state
b) Ans: DML, changes state, not schema

Q2. [3+3 Marks] There are two types of data independence in database management systems: Logical Data Independence, and Physical Data Independence. Given the scenarios below, Identify what type of data independence both are and Explain your answer briefly (at max 3 sentences) with reference to Three-Schema Architecture.

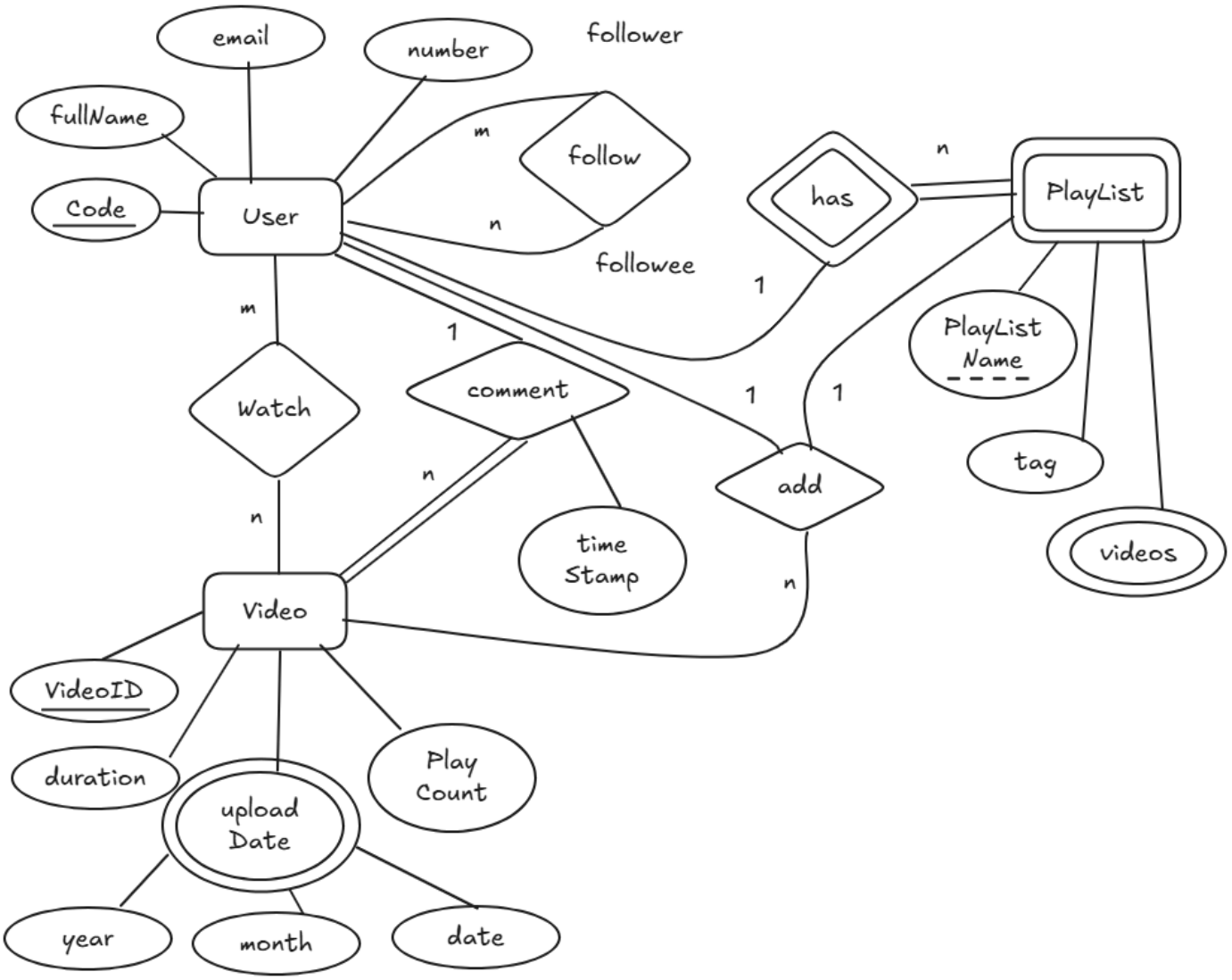
- You work for a social media company. The company has decided to move from cloud hosting to physical hardware to support the server due to increase in demand.
- You work for a social media company. The company has decided to track the number of hours a user has stayed on their platform. This tracked_hours is not visible to the user but kept in the database.

Ans: First Case, PDI, changes in internal schema that does not affect conceptual schema.
Second Case, LDI, changes in conceptual schema without affecting the External view.

Q3. [10 Marks] You are tasked with designing a Video Streaming System. The system must meet the following requirements:

- Each user will have a unique code, full name, email, and mobile number.
- Users can watch videos, which have a unique video ID, duration, upload_date, and play count.
- A video can be watched by many users and a user can watch many videos.
- Users can also comment on videos. Whenever a comment is made, the timestamp is recorded. A single user can comment multiple times.
- All videos must have comments, but not all users will leave a comment on the videos.
- Users can also follow other users.
- Some users might create a playlist. Each playlist has a playlist_name, tag, and a list of videos. Playlist_name is not unique, rather it is kept track of with the user's code and playlist_name combined.
- Users add videos to their playlist. One user can add many videos to their single playlist.

Now, draw an **ER diagram** of this system. Show at least 1 composite-multivalued attribute.
Write down all assumptions (if any).


email
number
follower
fullName
follow
m
n
Code
User
followee
has
n
PlayList
1
m
1
PlayList
Name
Watch
comment
1
1
n
add
tag
time
Stamp
videos
n
n
Video
VideoID
duration
Play
Count
upload
Date
year
month
date